Feature Engineering Log — HeavySuppliersWarehouseDatasets

Corrected 2026-10-28

Corrected version. Changes from the earlier log: (1) customer_features.csv monetary columns are labelled GST-inclusive, and GST-exclusive versions are added; (2) the snapshot date is stated; (3) the stock_ledger.csv columns added in validation are documented with their exact rule; (4) the P002 margin flag in product_features.csv is now False.

Row-level features added

sales_transactions.csv

- order_to_delivery_days — delivery_date minus order_date
- order_month, order_year — for time-series grouping
- invoice_payment_window_days — due_date minus invoice_date
- invoice_id_ambiguous — True on the 2,555 sales lines whose invoice shares its invoice_id with another invoice

purchase_transactions.csv

- po_planned_lead_days — expected_delivery_date minus order_date
- po_actual_lead_days — received_date minus order_date
- po_lead_variance_days — actual minus planned (positive = late)

New aggregate feature tables

customer_features.csv (500 customers, one row each) — RFM-style, built only from Delivered orders (cancelled orders don't count as sales activity). Snapshot date for recency and tenure: **2024-12-31** (the last order date).

Column	Definition
order_count	Number of Delivered orders (frequency)
total_revenue_incl_gst	Sum of line_grand_total over the customer's Delivered order lines (monetary, **including GST**). This is what the column total_revenue held in the earlier version.
total_revenue_ex_gst	Sum of line_total (**excluding GST**). New. Sums across all customers to 22,958,987,430, the same as the Net revenue 2019–24 baseline.
recency_days	Days from the customer's last Delivered order_date to 2024-12-31
avg_line_value_incl_gst	total_revenue_incl_gst divided by the number of Delivered order lines (earlier name: avg_line_value)
avg_line_value_ex_gst	total_revenue_ex_gst divided by the number of Delivered order lines. New.
customer_tenure_days	Days from customer_since to 2024-12-31

Note: the earlier columns total_revenue and avg_line_value were GST-inclusive but not labelled as such. They run about 27% above revenue excluding GST (29,253,993,162 vs 22,958,987,430 in total). Use the ex-GST columns when comparing with sales KPIs.

product_features.csv (30 products, one row each) — total_qty_sold / total_sales_revenue (from sales), total_qty_purchased / total_purchase_cost (from purchases), sell_through_ratio, gross_margin_realized, and avg_daily_qty_out — this last one is derived from stock_ledger OUT movements, deliberately NOT from inventory_master.current_stock, since that field is still flagged with the unresolved scale mismatch from the cleaning stage. (The validation stage added calculated_margin_percentage and margin_mismatch_flag. The stored P002 margin has been corrected to 60.0, a markup on cost, so the flag is False for all 30 products.)

Columns added to stock_ledger.csv (validation stage)

- **implied_signed_quantity** — the change in running_balance caused by the row, in file order within each product/branch. For the first movement of a product/branch the previous balance is inventory_master.opening_stock. For IN and OUT rows it equals +quantity / −quantity on all 234,776 rows. For ADJUSTMENT rows it is +quantity or −quantity (quantity itself is always stored positive).
- **adjustment_sign_ambiguous_flag** — True on all 2,454 ADJUSTMENT rows, meaning “use implied_signed_quantity, not quantity”.

Result: 1,227 adjustments decrease stock, 1,227 increase it, net +62 units. For every product/branch, opening_stock plus the sum of implied_signed_quantity equals the final running_balance.

Feature deliberately NOT built

A “below reorder point” flag (current_stock vs. reorder_level) was considered but dropped: since current_stock is above max_stock on every single row (100% of inventory_master), that comparison would never trigger meaningfully and would misrepresent real stock health. Use avg_daily_qty_out in product_features.csv instead for any velocity/reorder analysis until the scale issue is resolved with the source system.